AIR TRAFFIC ON TIME PERFORMANCE ANALYSIS

zoo un park

Project Overview

This document includes my project report on air traffic on-time performance analysis

using mySQL and R for visualization

Project

Nowadays, numerous flights are operating in a single day with various destinations and there are lots of unexpected cases where the flight schedule changes due to multiple reasons. The purpose of this report is to examine the dataset to solve various questions that can be questioned from air traffic-related topics and provide clear results to suggest what features need to be noted from airline on-time statistics. For this analysis, real-time data from 09/01/2019 to 09/30/2019 was used, with various information that will be used for different data analyses. On-time performance data contains records of flights by date, airline, originating airport, destination airport, and many other details related to flights. Topics that will be dealt with in this report include airport and airline departure analysis, weekly flight performance analysis, and cancellation reason analysis.

The first topic is airline and airport departure analysis. It is common for airlines to depart later than expected due to various reasons(i.e. weather), for this question, the first analysis for this is about the maximum departure delay in minutes for each airline.

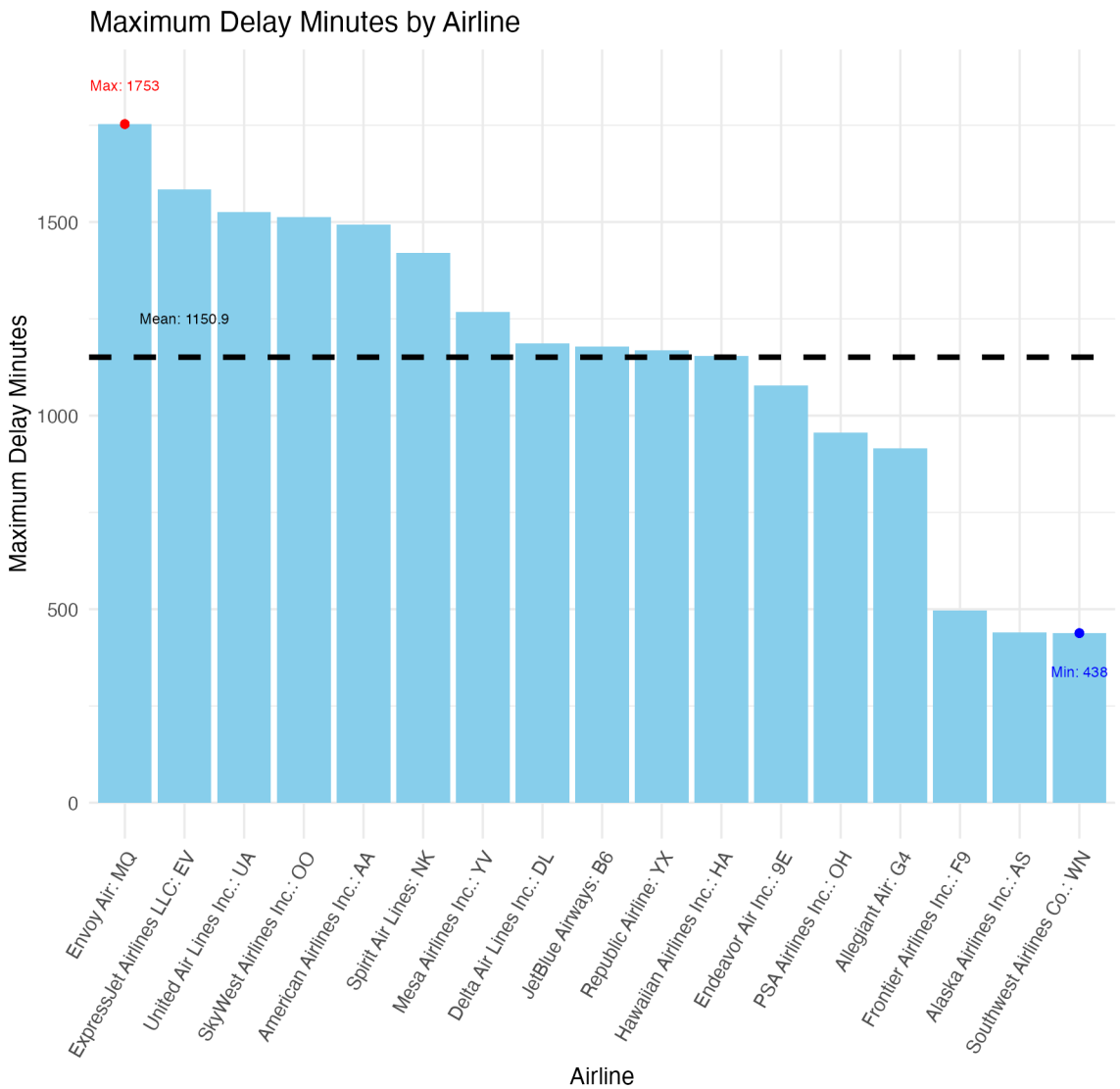


Figure1

figure 1 shows maximum delay minutes by airline order by longest to shortest maximum delay minutes in 09/01/2019 to 09/30/2019 timeframe. Envoy Air has the longest maximal delay minutes among all airlines with 1753 minutes delay and Southwest Airlines has the shortest delay minutes with 438 minutes and the mean of the maximum delay is 1150.9 minutes. Since this is a maximum delay not average, it has long delay minutes on average.

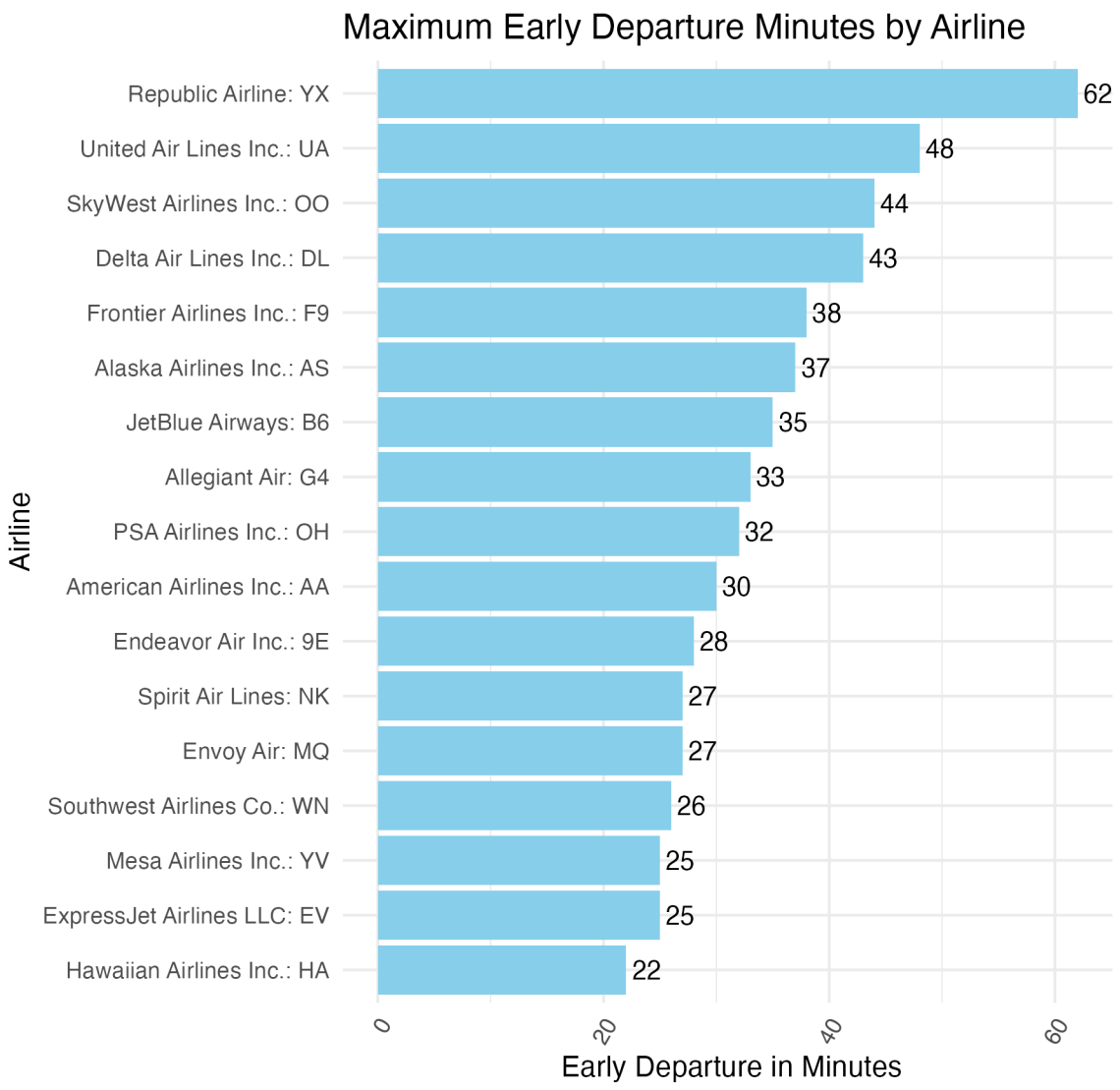


Figure2

The second analysis is on Maximum early departure in minutes by the airline. Figure 2 shows the maximal early departure minutes by airline. Overall Republic Airlines has maximum early departures of 62 minutes early departure, whereas Hawaiian Airlines has the least maximum early departure with 22 minutes early departure.

Table 1: Airport with Highest Average Departure Delay

Airport Name	Airport Code	Average Delay (minutes)
North Bend/Coos Bay, OR: Southwest Oregon Regional	13694	43.92

The third analysis is about the highest average airport departure delay. It is common in airports that flights are delayed due to various reasons. Table 1 shows the airport that has the highest average departure delay and Figure 4 shows this airport's location. From this data, North Bend/Coos Bay, OR: Southwest Oregon Regional Airport that is located in Oregon, USA has the highest average delay with approximately 43.92 minutes.

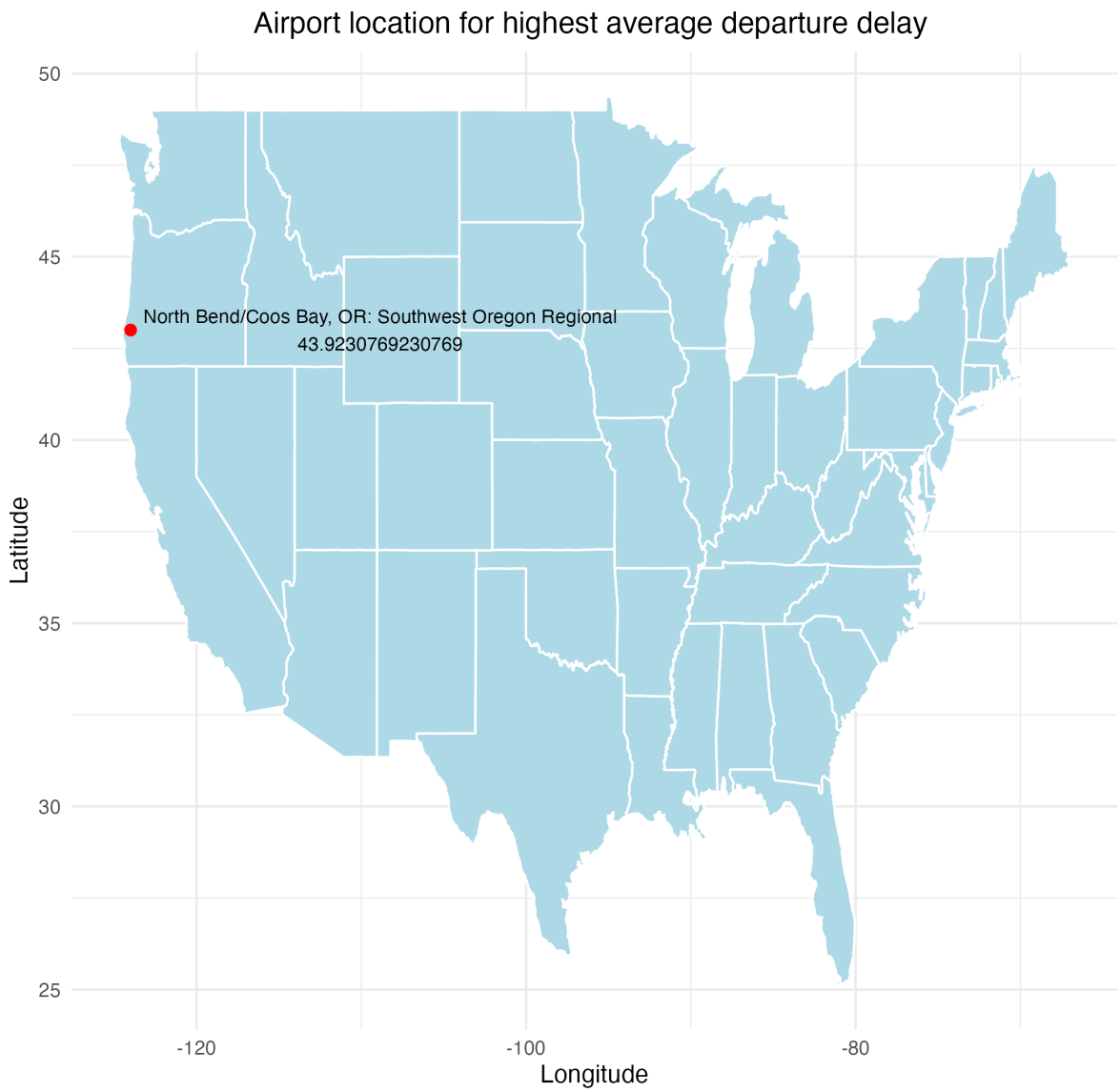


Figure4

The Fourth is finding airports with the highest average departure delayed airport for each airline. Figure 5 is a bar plot that shows airports that have the highest average departure delayed airport for each airline and Figure 6 shows the airport location with the highest average departure delays by airlines that corresponds with the bar plot. In Figure five most of the airline's maximum average delay by the airport is less than 100 minutes, but the top 2 airlines, Endeavor Air in General Mitchell International, Milwaukee, and Skywest airline in Philadelphia International, Philadelphia have quite an outlier, with 360.5 and 167 minutes delay each. Southwest has the lowest average departure delay in Newark Liberty International, Newark with 11.2 minutes, meaning that Southwest Airlines tends to depart accurately for all

airports.

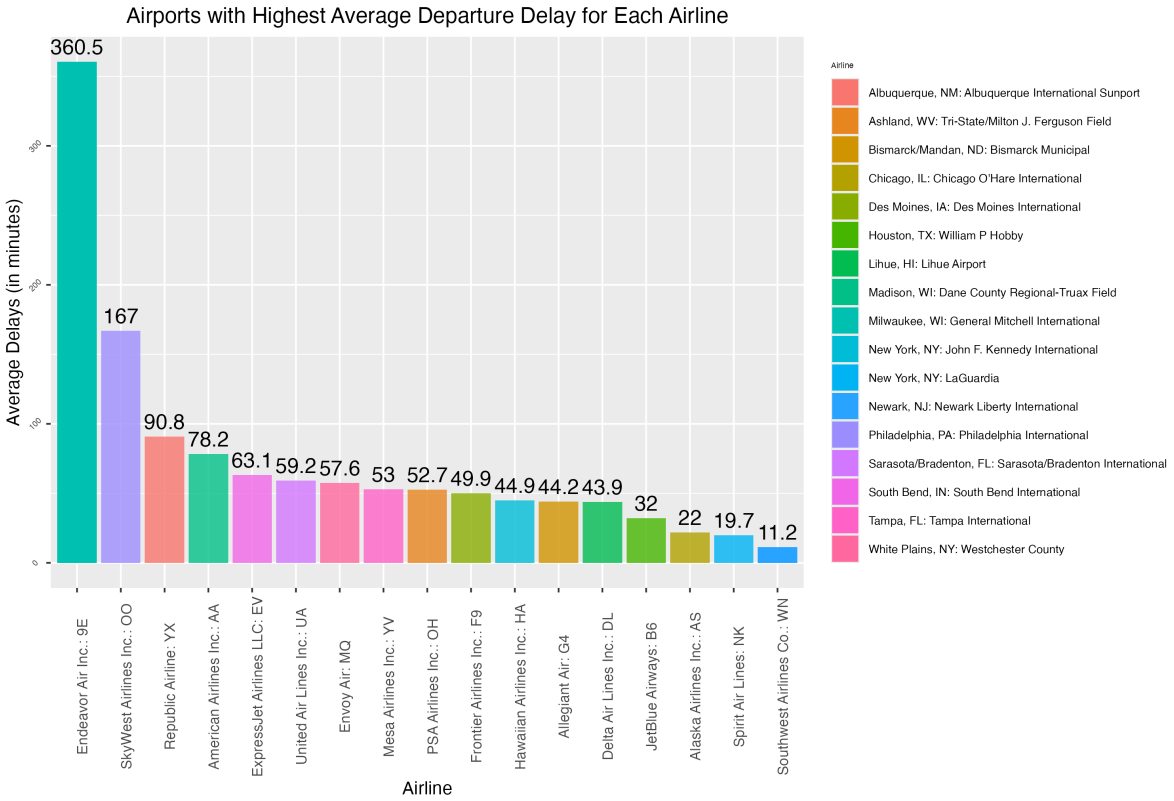


Figure5

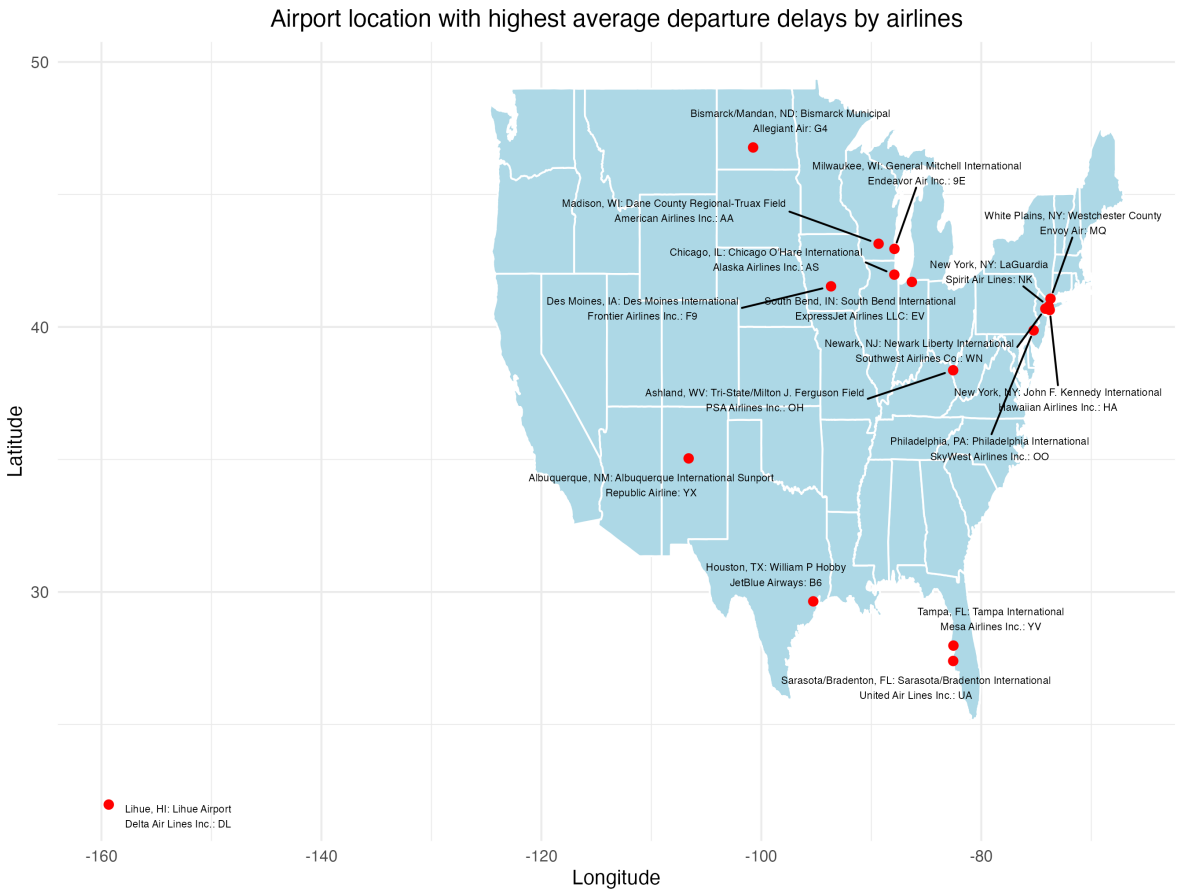
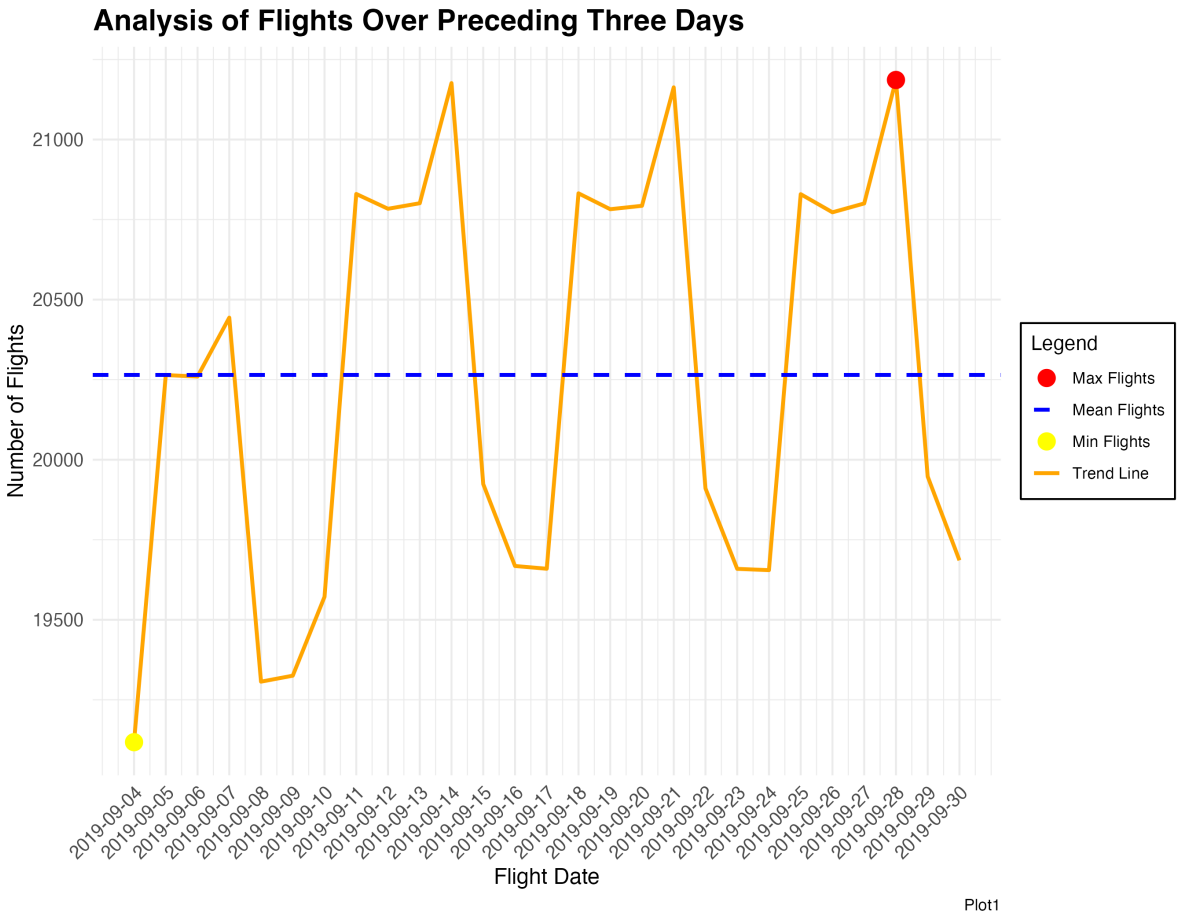


Figure6

Table 2: Rank of the Day in Week by Number of Flights Performed by All Airlines

Day of the Week	Number of Flights	Rank
Monday	106,461	1
Sunday	99,427	2
Thursday	85,135	3
Friday	85,095	4
Wednesday	81,677	5
Tuesday	81,150	6
Saturday	67,034	7

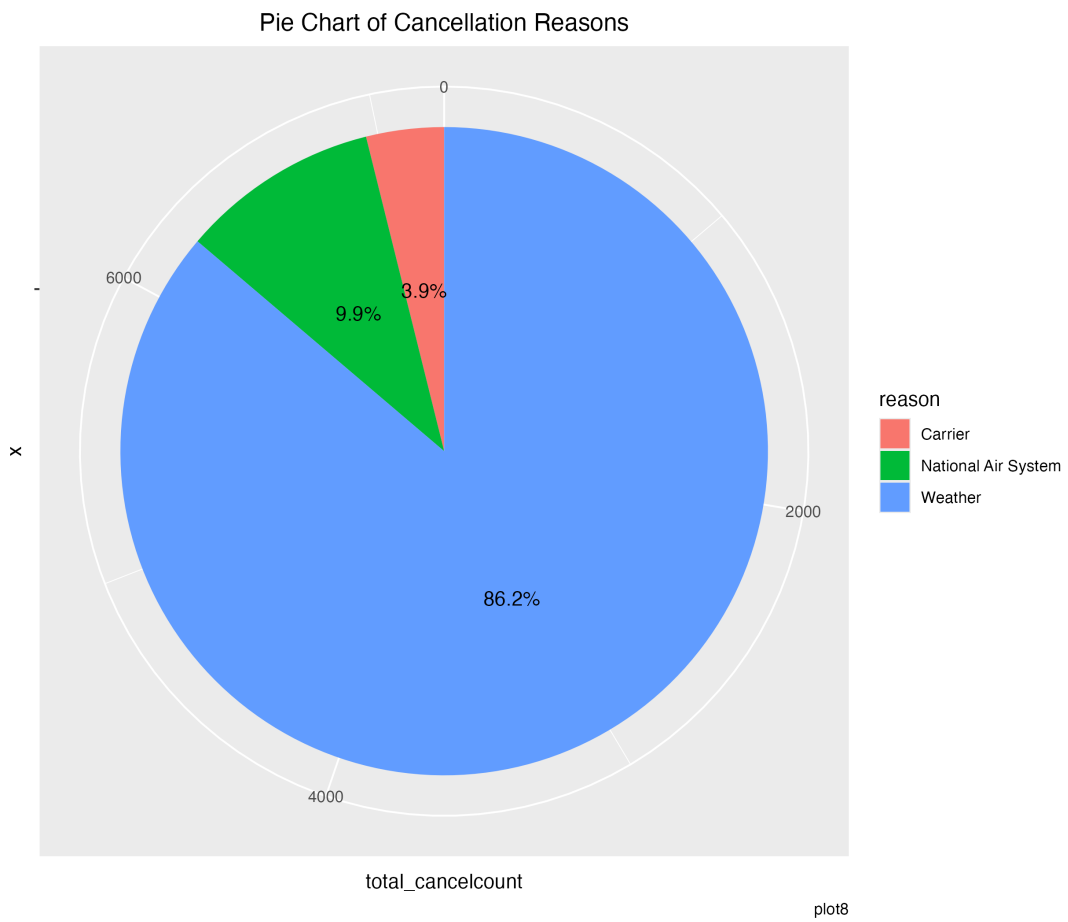
Mean: 87,711.29
Total: 605,979



The second topic is weekly flight performance analysis. The total number of flights operated could be different, some days might have higher demand, while other days might have less demand. Table 2 shows the day of the week's total number of flight performance and its rank. From the table, Monday and Sunday are the top 2 days that have the most flights operated, having more than 90,000 flights operated, while Thursday and Saturday are the bottom 2 days that have fewer flights operated. Plot 1 shows a line plot showing flights over three preceding days. In this line plot, there are the same trends for five-day intervals. Especially, 09/11,09/18,09/25 are the days that the number of flights cumulated over the past three days increases sharply than other dates. These days are Wednesday, which means it is a result of the sum of Sunday, Monday, and Tuesday flights. From Table 1, Monday and Sunday are the top 2 busiest days and this plot reflects this ranking. The third topic is cancellation reason analysis. There are cases in which instead of delays, flights are cancelled due to various causes. The first part of this analysis is to find how many flights are canceled and their proportion. Table 3 reflects the number of flights canceled and its percentage. From this data, 10,016 flights were canceled and its percentage is 1.7%, which could say it is low.

Table 3: Number of Flights Cancelled and Their Percentage

Flight Status	Number of Flights	Percentage
Cancelled	10,016	1.7%
Operated	595,964	98.3%



Plot 8 is a pie chart that shows the percentage of flight cancellation reasons for canceled flights during September 2019. weather condition takes most of the portion, having 86.2% for cancellation. This could be reasonable since September is usually regarded as a peak season for hurricanes in the USA, so this can be reflected in this reason. Other reasons for cancellation are national air systems, having 9.9% following after weather reasons and carrier reasons having the least portion, having 3.9%.

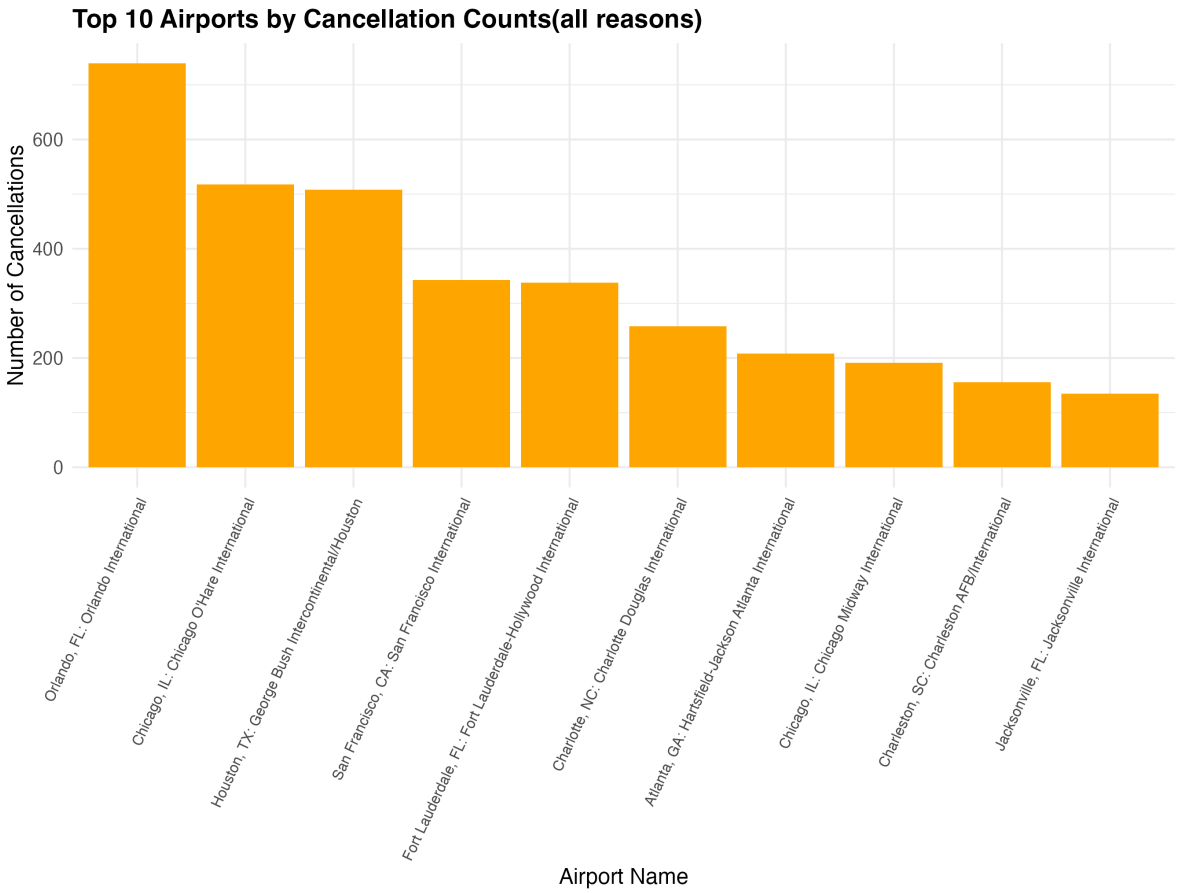


Figure8

figure8 shows the top 10 airports out of 285 airports that had cancellations regardless of reason. Airports in this plot are located in the southern part, which is quite influential to severe weather conditions during September, so this plot also supports that pie chart may have a reasonably high portion on weather reasons for cancellation.

R code for visualization

loading data

```{R}

library(dplyr)
d1<- read.csv("project_1.csv")
d2<- read.csv("project_2.csv")
```

```
d3<- read.csv("project_3.csv")
d4<- read.csv("project_4.csv")
d5<- read.csv("project_5.csv")
d6<- read.csv("project_6b.csv")
d7<- read.csv("project_7.csv")
```

figure1

```{R}
plot1<-ggplot(d1, aes(x = reorder(Airline, -MaxDelayMinutes), y = MaxDelayMinutes)) +
  geom_bar(stat = "identity", fill = "skyblue") +
  geom_point(aes(x = max_airline, y = max_delay), color = "red", size = 1) +
  # Max point
  geom_point(aes(x = min_airline, y = min_delay), color = "blue", size = 1) +
  # Min point
  geom_hline(yintercept = mean_delay, color = "black", linetype = "dashed", size = 1) +
  # Highlight mean
  labs(title = "Maximum Delay Minutes by Airline",
       x = "Airline",
       y = "Maximum Delay Minutes",
       caption = "Figure1") +
  annotate("text", x = max_airline, y = max_delay +100, label = paste("Max:", max_delay),
           color = "red",size = 2) +
  annotate("text", x = min_airline, y = min_delay - 100, label = paste("Min:", min_delay),
           color = "blue",size = 2) +
  annotate("text", x = 2, y = mean_delay + 100, label = paste("Mean:", round(mean_delay, 1))
           color = "black", hjust = 0.5,size = 2) +
  theme_minimal(base_size = 9) +
  theme(axis.text.x = element_text(angle = 60, hjust = 1),
        plot.background = element_rect(fill = "white", color = NA) )
ggsave("my_plot1.png", plot = plot1, width = 6, height = 6)

```

figure 2

```{R}
summary_data <- d2 %>%
  mutate(EarlyDepartureMinutes = abs(EarlyDepartureMinutes)) %>%
  summarise(
    Max = max(EarlyDepartureMinutes),
    Min = min(EarlyDepartureMinutes),
    Mean = mean(EarlyDepartureMinutes)
  )

plot2b<-ggplot(d2, aes(x = reorder(Airline, abs(EarlyDepartureMinutes)),
 y = abs(EarlyDepartureMinutes))) +
  geom_bar(stat = "identity", fill = "skyblue") +
  coord_flip() +
  geom_text(aes(label = abs(EarlyDepartureMinutes)), hjust = -0.2, size = 3.5) +
  labs(title = "Maximum Early Departure Minutes by Airline",
       x = "Airline",
       y = " Early Departure in Minutes",caption = "Figure2") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 60,
   hjust = 1),
        plot.background = element_rect(fill = "white",
        color = NA) )
ggsave("my_plot2b.png", plot = plot2b, width = 6, height = 6)

```

figure 6

```{R}
library(maps)
library(ggrepel)
a11<-read.csv("/Users/zp/Desktop/a111.csv")
lat1<- 43
long1<- -124
air1<- d4
air1$Latitude<- lat1
air1$Longitude<-long1
```

```
usa_map <- map_data("state")
plot4<-ggplot() +
  # Draw the USA map
  geom_polygon(data = usa_map, aes(x = long, y = lat, group = group),
               fill = "lightblue", color = "white") +

  # Plot all airports as points
  geom_point(data = air1, aes(x = Longitude, y = Latitude),
             color = "red", size = 2) +

  # Add labels for all airports
  geom_text_repel(data = air1,
                  aes(x = Longitude, y = Latitude,
                      label = paste(AirportName,AvgDelay, sep = "\n")),
                  size = 3, box.padding = 0.5, point.padding = 0.5,
                  max.overlaps = 50) + # Adjust overlaps limit

  # Titles and themes
  theme_minimal() +
  labs(title = "Airport location for highest average departure delay ",
       x = "Longitude", y = "Latitude",caption = "Figure4") +
  theme(plot.title = element_text(hjust = 0.5),
  plot.background = element_rect(fill = "white", color = NA))
ggsave("my_plot4.png", plot = plot4, width = 7, height = 7)
```

figure 6

```{R}
library(ggplot2)
library(ggrepel)
library(maps)
library(usmap)
latitude<-a11$Latitude
longitude<-a11$Longitude
airport<- d5
airport$latitude<-latitude
airport$longitude<-longitude
# Plot with all airports and labels
plot5<-ggplot() +
```

```
  # Draw the USA map
  geom_polygon(data = usa_map, aes(x = long, y = lat, group = group),
               fill = "lightblue", color = "white") +

  # Plot all airports as points
  geom_point(data = airport, aes(x = longitude, y = latitude),
             color = "red", size = 2) +

  # Add labels for all airports
  geom_text_repel(data = airport,
                  aes(x = longitude, y = latitude,
                      label = paste(AirportName, AirlineName, sep = "\n")),
                  size = 2, box.padding = 0.5, point.padding = 0.5,
                  max.overlaps = 50) + # Adjust overlaps limit

  # Titles and themes
  theme_minimal() +
  labs(title = "Airport location with highest average departure delays by airlines",
       x = "Longitude", y = "Latitude",caption = "Figure6") +
  theme(plot.title = element_text(hjust = 0.5),
  plot.background = element_rect(fill = "white", color = NA))
ggsave("my_plot5.png", plot = plot5, width = 9, height = 7)


```
```
```

```
  )

# Plot
plot5c<-ggplot(d5, aes(x = reorder(AirlineName, -AvgDelay),
y = AvgDelay, fill = AirportName)) +
  geom_bar(stat = "identity", aes(alpha = Highlight)) +
  geom_text(aes(label = round(AvgDelay, 1)),vjust = -0.5, size = 4.5) +
  #coord_flip() +
  scale_alpha_manual(
    values = c("Max" = 1, "Min" = 1, "Normal" = 0.8),
    guide = "none"
  ) +
  labs(
    title = "Airports with Highest Average Departure Delay for Each Airline",
    x = "Airline",
    y = "Average Delays (in minutes)",
    fill = "Airline",caption = "Figure5"
  ) +
  theme(
    axis.text.x = element_text(angle = 90, hjust = 0.5),
    axis.text.y = element_text(angle = 45.,hjust = 0.5,size = 5),
    plot.title = element_text(hjust = 0.5),
    plot.background = element_rect(fill = "white", color = NA,size = 10),
    legend.text = element_text(size = 7),
    legend.title = element_text(size = 5)
  )

#  theme(plot.title = element_text(hjust = 0.5),
#plot.background = element_rect(fill = "white", color = NA))
ggsave("my_plot5c.png", plot = plot5c, width = 10, height = 7)

```
```
```
```{R}
overall_reasons <- most_frequent_reasons %>%
  group_by(reason) %>%
  summarise(total_cancelcount = sum(total_cancelcount))
```

figure 8

```
# Calculating percentages
overall_reasons <- overall_reasons %>%
  mutate(percentage = total_cancelcount / sum(total_cancelcount) * 100)

# Creating the pie chart with percentages
plot6<-ggplot(overall_reasons, aes(x = "", y = total_cancelcount, fill = reason)) +
  geom_bar(width = 1, stat = "identity") +
  coord_polar("y", start = 0) +
  geom_text(aes(label = paste0(round(percentage, 1), "%")),
            position = position_stack(vjust = 0.5)) +
  labs(title = "Pie Chart of Cancellation Reasons",caption = "Figure7") +
  theme(plot.title = element_text(hjust = 0.5),
  plot.background = element_rect(fill = "white", color = NA))
ggsave("my_plot6.png", plot = plot6, width = 9, height = 7)
```

plot 8

```{R}
top_cancellation_airports <- d6 %>%
  group_by(AirportName) %>%
  summarise(cancelcount = sum(cancelcount)) %>%
  arrange(desc(cancelcount)) %>%
  slice(1:10)

plot6b<- ggplot(top_cancellation_airports,
 aes(x = reorder(AirportName, -cancelcount), y = cancelcount)) +
  geom_bar(stat = "identity", fill = "orange") +
  labs(
    title = "Top 10 Airports by Cancellation Counts(all reasons)",
    x = "Airport Name",
    y = "Number of Cancellations",caption = "Figure8"
  ) +
  theme_minimal() +
  theme(

    plot.background = element_rect(fill = "white", color = NA),  # White background
    axis.text.x = element_text(angle = 65, hjust = 1, size = 8),  # Rotate x-axis text
    axis.text.y = element_text(size = 10),  # Adjust y-axis text size
    axis.title = element_text(size = 12),  # Adjust axis title size
```

```
    plot.title = element_text(size = 14, face = "bold")  # Larger bold title
  )

ggsave("my_plot6b.png", plot = plot6b, width = 9, height = 7)
```

plot 1

```{R}
# Load necessary libraries
library(ggplot2)
d7$FlightDate <- as.Date(d7$FlightDate)
# Find max and min flights
max_flights <- d7[which.max(d7$flights_over_preceding_three_days), ]
min_flights <- d7[which.min(d7$flights_over_preceding_three_days), ]
mean_flights <- mean(d7$flights_over_preceding_three_days)
# Create the plot
plot7<-ggplot(d7, aes(x = FlightDate, y = flights_over_preceding_three_days)) +
  geom_line(aes(color = "Trend Line"), size = 1) +  # Add trend line
  geom_point(data = max_flights,
  aes(x = FlightDate, y = flights_over_preceding_three_days,
   color = "Max Flights"), size = 4) +  # Highlight max
  geom_point(data = min_flights,
  aes(x = FlightDate, y = flights_over_preceding_three_days,
  color = "Min Flights"), size = 4) +  # Highlight min
  geom_hline(aes(yintercept = mean_flights,
   color = "Mean Flights"), linetype = "dashed", size = 1) +
    # Add mean line
  scale_color_manual(values = c("Trend Line" = "orange",

                                "Max Flights" = "red",
                                "Min Flights" = "yellow",
                                "Mean Flights" = "blue")) +
                                 # Define custom colors
  labs(title = "Analysis of Flights Over Preceding Three Days",
       x = "Flight Date",
       y = "Number of Flights",
       color = "Legend",caption = "Plot1") +  # Add labels
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))+
```

```
  scale_x_date(breaks = seq(min(d7$FlightDate), max(d7$FlightDate), by = "1 day"))+
  theme(plot.title = element_text(size = 16, face = "bold"),
        plot.background = element_rect(fill = "white", color = NA),
        axis.title = element_text(size = 12),
        axis.text = element_text(size = 10),
        legend.background = element_rect(fill = "white", color = "black"))

ggsave("my_plot7.png", plot = plot7, width = 9, height = 7)

```
```
```

mySQL Code

```
```{sql}
#1
SELECT
    Name AS Airline,
    MAX(DepDelay) AS MaxDelayMinutes
FROM
    al_perf
JOIN
    L_AIRLINE_ID
ON
   al_perf.DOT_ID_Reporting_Airline =   L_AIRLINE_ID.ID
GROUP BY
    Name
ORDER BY
    MaxDelayMinutes desc;

#17 rows returned
# from al_perf, select Name and set as airline,select maximum number of depDelay
#since i am finding maximal departure.  set this name as
# MaxDelayMinutes
#then join with L_AIRLINE_ID
#with condition al_perf's DOT_ID_Reporting_Airline equals with L_AIRLINE_ID's ID
#then group by Name
#lastly order it by EarlyDEpartureMinutes descending

#2.
SELECT
```
```

```
    Name AS Airline,
    min(DepDelay) AS EarlyDepartureMinutes
FROM
    al_perf
JOIN
    L_AIRLINE_ID
ON
   al_perf.DOT_ID_Reporting_Airline =   L_AIRLINE_ID.ID
GROUP BY
    Name
ORDER BY
    EarlyDepartureMinutes ASC;

#17 rows returned
# from al_perf, seect Name and set as airline,select minimum number of depDElay
#since i am finding early departure, that is with negative value, set this name as
# EarlyDeparttureMinutes
#then join with L_AIRLINE_ID
#with condition al_perf's DOT_ID_Reporting_Airline equals with L_AIRLINE_ID's ID
#then group by Name
#lastly order it by EarlyDEpartureMinutes accending(negative values)


#3.
SELECT
    L_WEEKDAYS.Day AS Day,
    COUNT(*) AS NumberOfFlights,
    rank() OVER (ORDER BY COUNT(*) DESC) as Ranking

FROM al_perf
JOIN L_WEEKDAYS  on al_perf.DayOfWeek = L_WEEKDAYS.Code
GROUP BY
    DayOfWeek
ORDER BY
    NumberOfFlights DESC;

#7 rows returned
# select days in L-WEEKDAYS and set as Day,
#count all and set as NumberOfFlights
#and use rank() over to rank the NUmberOfFlights
#as set previous step and set this as ranking
#these are from al_perf(Day from L_WEEKDAYS)
```

```
#now join with L_WEEKDAYS with condition
#al_perf's DayOfWeek is equal to L_WEEKDAYS's Code
#next group it by DayofWeek
#finally order it respect to NumberOfFlights


#4.
WITH AirportDelays AS (
    SELECT
        s.OriginAirportID AS AirportID,
        AVG(s.DepDelayMinutes) AS AvgDelay
    FROM
        al_perf s
    GROUP BY
        s.OriginAirportID
)
SELECT
    a.Name AS AirportName,
    a.ID AS AirportCode,
    d.AvgDelay
FROM
    AirportDelays d
JOIN
    L_AIRPORT_ID a
ON
    d.AirportID = a.ID
ORDER BY
    d.AvgDelay DESC
LIMIT 1;
#1 row returned
# create temporary relation call as airprotDelays using with, inside,
#select OriginAirportID,set as AirportID and
#averaged DepDelayMinutes set as Avgdelay
#DePDelayMInutes set 0 for early departures,
#so it considers departure delays and 0 minute delay
#all these from al_perf, group it by AirportID
#next, select name and set as AirportName, ID ,
#set as AirportCode,  AvgDelay from AirportDelays
#join this with L_AIRPORT_ID with condition AirportID
# in AirportDelays equals to L_AIRPORT_ID's ID
#order it repsect to AvgDelay
#finally, output just one line, using limit 1
```

```
#5.
WITH airlineairportdelay AS (
    SELECT
        s.DOT_ID_Reporting_Airline AS AirlineID,
        s.OriginAirportID AS AirportID,
        AVG(s.DepDelayMinutes) AS AvgDelay
    FROM
        al_perf s
    GROUP BY
        s.DOT_ID_Reporting_Airline, s.OriginAirportID
),
maxdelayperairline AS (
    SELECT d.AirlineID,d.AirportID,d.AvgDelay
    FROM airlineairportdelay d
    JOIN (
        SELECT
            AirlineID,MAX(AvgDelay) AS MaxDelay
        FROM airlineairportdelay
        GROUP BY AirlineID
    ) m
    ON d.AirlineID = m.AirlineID AND d.AvgDelay = m.MaxDelay
)

SELECT a.Name AS AirlineName,p.Name AS AirportName,m.AvgDelay
FROM maxdelayperairline m
JOIN L_AIRLINE_ID a ON m.AirlineID = a.ID
JOIN L_AIRPORT_ID p ON m.AirportID = p.ID;

#17rows returned
#create temporary relation name as AirlineAirportDelay
#inside,  select DOT_D_reporting_Airline name as airlineID
#OriginAirportID name as AirportID
#averaged DepDelayMinutes name as
#avgdelay(DepDelayMinuutes set 0 for early departure so it is much
#better to use this rather than using DepDelay )
#now group it by DOT_D_reporting_Airline name,OriginAirportID
#create temporary relation maxdelayperairline
#inside, first select AirlineID,AirportID,
#AvgDelay from airlineairportdelay
#join with m, where m is a table that
```

```
#select AirlineID,maximum of AvgDelay that i got from
#airlineairportdelay,name as MaxDelay
#these are from airlineairportdelay, then group it by AirlineID,
#this is expected to give maximum average delay for each AirlineID
#join airlineairportdelay with m,
#with condition airlineairportdelay and m's AirlineID is equal
#and airlineairportdelay's AvgDelay and m's MaxDelay is equal
# select Name from L_AIRLINE_D, avgDelay from
maxdelatperairline(one got just before)
#Name from L_AIRPORT_ID name as AirportName
#for join with L_AIRLINE_ID with
condition AirlineID from m and ID from L_AIRLINE_ID is equal
# join again with L_AIRPORT_ID with
condition AirportID form m and ID from L_AIRPORT_ID is equal

#6a
select * from al_perf
where Cancelled = 1;
#10016 rows returned
# select all from al_perf, with condiion Cancelled = 1, since
#Cancelled =1 refers to flight has been cancelled

#6b
WITH cancel AS (
    SELECT
        s.OriginAirportID AS airportID,
        c.Reason AS reason,
        COUNT(*) AS cancelcount
    FROM
        al_perf s
    JOIN
        L_CANCELATION c
    ON
        s.CancellationCode = c.Code
    WHERE
        s.Cancelled = 1
    GROUP BY
        s.OriginAirportID, c.Reason
),
frequentreason AS (
    SELECT
        airportID,
```

```
        reason,
        cancelcount
    FROM
        cancel
    WHERE
        cancelcount = (
            SELECT
                MAX(cancelcount)
            FROM
                cancel cc
            WHERE
                cc.airportID = cancel.airportID
        )
)
SELECT
    a.Name AS AirportName,
    m.reason,
    m.cancelcount
FROM
    frequentreason m
JOIN
    L_AIRPORT_ID a
ON
    m.airportID = a.ID;
#285 rows returned
# create temporary relation using with naming as cancel
#inside, select OriginAirportID from al_perf,name as airportID
#Reason from L_CANCELLATION, count all naming as cancelcount
#join al_perf with L_CANCELATION with condition CancellationCode in al_perf equals to
#L_CANELATION'S Code. also Cancelled in al_perf is 1(1 if flight cancelled)
#then group by OriginAirportID and Reason
#next set temporary relation naming as frequentreason
##select airportID,reason,cancelcount from cancel(one created before)
#with condition on cancelcount; select maximum
#of concelcount from cancel name as cc(one created before)
#with condition airportID in cc is equal to airprotID in cancel
#finally selct Name from L_AIRPORT-ID, reason
#and cancelcount from frequentreason(one created just before)
# join with L_AIRPORT-ID  with condition airportID
# in frequentreason is equal to L_AIRPORT_ID'S ID
```

```
#7
WITH countflight AS (
    SELECT FlightDate,COUNT(*) AS dayflights
    FROM al_perf
    GROUP BY FlightDate
),
averageflightbyday AS (
    SELECT c1.FlightDate,
        CASE
            WHEN COUNT(c1.dayflights) < 3 THEN NULL
            ELSE AVG(c2.dayflights)
        END AS flights_over_preceding_three_days
    FROM countflight c1
    LEFT JOIN countflight c2
    ON c2.FlightDate BETWEEN DATE_SUB(c1.FlightDate, INTERVAL 3 DAY)
    AND DATE_SUB(c1.FlightDate, INTERVAL 1 DAY)
    GROUP BY c1.FlightDate
)
SELECT FlightDate,flights_over_preceding_three_days
FROM averageflightbyday
WHERE flights_over_preceding_three_days IS NOT NULL
ORDER BY FlightDate;
#27 rows returned
# create temporary relation name as countflight
#inside flightDate and count it all, name this as dayflights
#from al_perf and group it by FlightDate
# create temporary relation name as averageflightbyday
#inside create condition for first three days, set as null because
#information needed average number of flight over the 3 proceeding day
#so for first three days, there would be no average number of flights
# select  flightDate from countflight name as c1
# now left join with countflight that name as c2
# with condition; here i am finding average flights for three day interval
#so by using DATE_SUb, i am subtracting three days from FlightDate,
#which will expected to show start of the range
#for second part in between~and, this will perform same, but subtracting
#one day from flightDate
#so this line will give date range needed for three proceeding date
#which is range 1-3 days for computaion on number of flights over the 3 proceeding date
#next, group it by FlightDate
# select FlightDate,flights_over_preceding_three_days from averageflightbyday
# put condition only produce output that is not null, that is
```

```
#exclude first three days(in my case 2019/09/01- 2019/09/03 will be excluded)
#finally, order it by FlightDate



```